

PONGCHULA PEMACHOLING TSHARZAW TSHOGPA MANGEMENT AND MARKETING PLAN

(FIRST REVISION)



Phangkhar Gewog, Panbang Dungkhag Zhemgang Dzongkhag

Plan Period: 2023 to 2027

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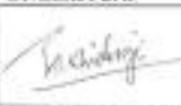
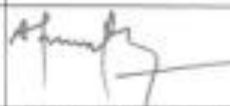
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APPROVAL SHEET

APPROVAL SHEET

Name of NWFPs Management and Marketing Group:	Pongchula Pomecholing Tsharzew Tshoppa
Dzongkhag:	Zhemgang
Gewog:	Phangkhaz
Village(s):	Gujong, Redjong, Pongchula
No. of households:	16 H/H
Area (ha)	50.22
Plan Period:	5 years

Prepared and submitted for approval:

	Chairperson NWFP Management & Marketing Group	Gap of the Gewog	Head, Social Forestry Section of Division/Park	Chief Forestry Officer of Division/Park
Signature:				
Name:	Tshering Yeshi	Tashi	Tashi Dorji PFO	Samten Wangchuk
Date:				

REVIEWED AND RECOMMENDED FOR APPROVAL:

	Reviewing Officer, FRPMD	Chief forestry Officer, FRPMD	Head of the Department
Signature:			
Name:	SONAM PELDON	SONAM TOBGAY	
Date:	01/04/2023	07/04/2023	

Approved by:


SECRETARY
 Ministry of Energy and Natural Resources



འབྲུག་རྒྱལ་ཁབ་ཀྱི་ རྒྱལ་ཁབ་དང་རང་བཞིན་ལོན་ཏེ་ཚད་ཀྱི་ འཕུལ་ཚབ་དང་རྒྱུ་ཁུངས་
ཤྱི་ལས་ཁུངས་

ROYAL GOVERNMENT OF BHUTAN
MINISTRY OF ENERGY & NATURAL RESOURCES
DEPARTMENT OF FORESTS & PARK SERVICES



DoFPS/FRPMD/NWFP-01/2023/ 895
NOTESHEET

April 7, 2023

Subject: Approval of NWFP Management Plans

The Non-Wood Forest Products are an important source of livelihood for many rural communities in the country. In order to ensure the sustainable management of resources in the wild, management responsibilities are handed over to the communities through the formation of Non-wood Forest Product Management and Marketing Group. Further, it is mandatory for the group to prepare a resource management and marketing plan and submit it for approval through the Chief Forestry Officer of that area. As such the Forest Resource Planning and Management Division had received and reviewed the following plans submitted and hereby submitted to seek approval:

1. Panabe Bamboo Management Group, Phangkhar Gewog, Zhemgang
2. Pongchula Pemacholing Tsharzew Tshogpa, Phangkhar Gewog, Zhemgang
3. Jomolhari Ngomoen Tshogpa, Soe Gewog, Thimphu
4. Yaksa Ngomoen Tshogpa, Tsento Gewog, Paro
5. Lengkhar Tsoy Wonghab Tshogpa, Khamdang Gewog, Tashiyangtse

As per the Forest and Nature Conservation Act of Bhutan, 1995, such plans are required to be approved by the head of the Ministry. Submitted for kind approval.

Submitted by:



(Souren Tenzin)
Chief Forestry Officer

Remarks:



(Lobsang Dorji)
DIRECTOR

Remarks: *Approved as proposed.*



Offg. SECRETARY

1. Introduction

The Pongchula Pemacholing Tsharzew Tshogpa falls under Phangkhar geog, Zhemgang district. In the previous terms, there were 25H/Hs as members but currently, with this renewal of the plan, the Members have dropped to 16H/H due to very unavoidable circumstances like death and migration to urban areas. The main aim of this program is to protect and make additional income through the sale of Bamboo handicrafts and other Non-Wood Forest Products (NWFPs) like Bamboo shoots and mushrooms etc. Such institution development felt necessary for the revival of the traditional practices of the kheng people and was accordingly approved by the Gewog Tshogdhu(GT). This was 2nd term of the plan for the people of Pongchula chiwog. The Forestry staff from Gomphu Range have facilitated in reviewing of this NWFP management plan, under the guidance of RMNP, Gelephu.

The raw materials identified for the Weaving of the handicraft's products are scattered around the village settlements. The resources will be collected sustainably based on the harvesting guidelines prescribed in the plan drawn in consultation with the management group members. To ensure effective management of resources and smooth functioning of the group, simple by-Laws were framed by the members, in line with the FNCR.

In the previous plan, they generated around nu 74,800/- from the sale of Handicraft products despite the covid pandemic situation. The marketing of the finished products will be done by the executive members of the group. The possible markets like; Panbang, Gelephu, Thimphu, and, during the time of special events, around the country. It should be the responsibility of the group members to prepare the products at any time and contribute to sell the products through one outlet with uniform prices for the products.

This plan is prepared mainly to facilitate the collection of resources from the forest very prudently by the members. The plan would empower and also guide the user group to implement activities in line with the management plan. The plan is valid for 5 years and within this term, the group would use the resources very sustainably to uplift their annual income.

2. Goals & Objectives

- To generate an income for the people of the Pongchula community from the sale of Tsharzew(bamboo handicraft) production.
- To ensure the sustainable harvesting of the Cane and Bamboo resources
- To promote indigenous bamboo handicrafts diversity

3. Description of NWFPs

Table 1: Brief description of NWFPs for commercial harvesting

NWFP name	Harvesting area(s) (name)	Parts harvested	Use purposes (Traditional & others)	Harvesting season (months)	The estimated quantity that can be collected by the group in a year
Bamboo	Shingmarpong, Rendi bee, Yungba, Virang toe,	Shoot	Vegetable items	July-sept	100kg
		Culm	Constructions and Chadree purposes.	December – February	3000 Culm/yr
Cane	Dhimla, Redibrang, Charsapong, Kharila	culm	handicrafts	Jan-Dec	50backload/year
Mushroom	Gujong, kradijong, cha pgon	fungus	Vegetable	Jan-march-sept-oct	100pkts
Yula (Neomicrocalamus Andropogonifolius)	Manling, Chapgon, redibrang, charsapong, dhimla, Roksar,	culm	Handicraft items	Dec-march	1500culms

4. Harvesting prescriptions for sustainable NWFP collection

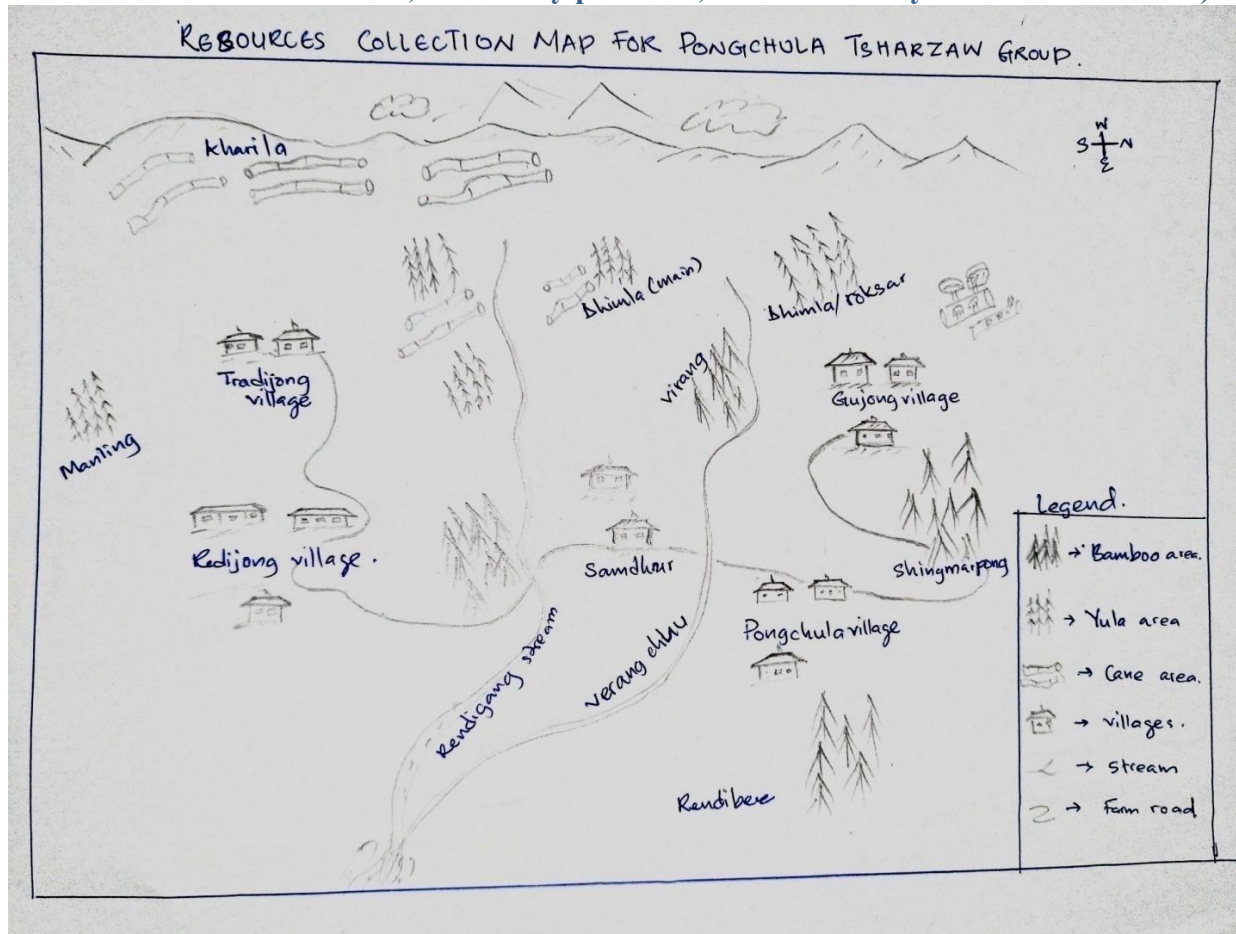
To make the harvesting of NWFP sustainable, the Pongchula NWFP group has to apply an appropriate harvesting prescription based on scientific and local knowledge. The chairman will work according to the seasonal calendar and the period of the guidelines. During the harvesting season, the chairman will collect a permit from the Gomphu Range office for the required head loads in the name of the NWFP group *and* then inform the members to proceed with harvesting the resources. The harvesting areas will be identified to the members as per the above table No. 1. The monitoring works will be also done by the committee members for the proper harvesting as per the guidelines prescribed hereunder.

Table 2: Harvesting prescriptions for sustainable NWFP harvesting

NWFP species	Harvesting prescriptions for sustainable harvesting
D. hamiltonii /pagshii (Bamboo)	For culms • Cut culms above the 1 node from the ground level and leave the culm below one year • Retain 50% of the healthy culms from the clump of different age • Harvest the culm in a very condensed or congested clump in the "U" shape manner. For shoots • Collect bamboo shoots of not more than one foot and leave about 75% of the shoots from the clump.

	• Cut the shoots from the ground level without damaging the buds and cover them with soil • Cut bamboo shoots with a sharp knife at ground level without damaging healthy ones. Cover the place with topsoil • Protect the remaining shoots
Mushroom	• Do not dig out the whole fungus from the wood log. • Pluck it out carefully • Retain the younger ones (as per the traditional knowledge base).
Yula(<i>Neomicrocalamus Andropogonifolius</i>)	▪ Harvesting should be limited to one, two-year-old or older culms (when culms produce a second shoot (branch) at the top); ▪ Harvest culms only from clumps that have more than five culms; ▪ Two to three young (one to two-year-old) culms should be maintained in each clump to favor more shoot regeneration in the following season; ▪ Culms should be harvested between November and May (the growth of Yula shoots is hampered if younger culms are damaged. Therefore, the Yula area should not be visited during the rest of the year). ▪ Livestock should be kept out of the growing areas of Yula, especially during the period from May to November; the presence of cattle will stunt the growth of young shoots.
Cane (calamus spp)	▪ Harvest canes when mature. The maturity of the cane can be recognized by several features: ○ loosening of leaf sheath/ peeling off of the “bark” of the cane ○ exposure of the lowest part of the stem ○ the cane colour turns yellowish-reddish along the entire cane length ○ an estimated cane length of 15 m (or a usable length of about 10 m) ▪ Harvest during winter as canes have a better quality during this season (lower water content and more durable). ▪ Do not harvest over-matured canes of <i>Calamus acanthospathus</i> as they are not suitable for handicraft purposes; leave these canes for the production of seeds. ▪ Harvest over-matured canes of <i>Plectocomia Himalayan</i> and <i>Calamus latifolius</i> to stimulate the regeneration of new shoots. ▪ Select about 10 cane plants (including both male and female plants) of the same species to be retained as seed canes per acre.

5. Sketch map of the NWFP (mainly Yula and Bamboo) harvesting areas (based on the traditional boundaries, customary practices, and availability of NWFP resources).



6. Google map of Pongchula showing the proposed Bamboo and Yula collection site



7. Walking distance of respective hamlets in reaching the NWFP collection site in hours:

Sl.No:	Name of hamlet	Collection area	Area (ac.)	Walking Distance (hour)
1	Gujong, Pongchula Redijong, Chapgon, Chapgon	Roksar	5.25	30min
2		Dhimla	28.8	5hrs
3		Virangang tooth	16	1hr
5		Yungbaleng	5.41	30mins
6		shingmarpong	18.7	20mins
7		Rendhibee	1	1:30mins
8		Charsapong	6.85	2hr
9		Redibrang	16.2	3hr
10		Kharila	23.3	5hrs
11		Manling	2.55	1hr

8. Marketing and sales plan

In the earlier plan period, the Pongchula Tsharzew group generated around Nu.74800/- from the sale of their Tsharzew products and so at the end of the plan period they divided the income equally among themselves.

The following products listed will be the main products that will be sold in the market base on the demand made by the customers. The sale plan was formulated based on the interim framework for

management and marketing of the NWFP plan book 2011. The prices are fixed based on the prevailing local market rates as tabulated.

Table 3: Marketing and sales plan for other NWFP

NWFP	Potential market	The estimated quantity for sale per season (unit)	Estimated price per unit(Nu)	Total (Nu)
Bamboo (shoot)	Gelephu market, pantang & other construction companies,KNC(Khhengrig Namsum Cooperatives),goling IFPP,nearby indian boarder city (Barbeta bazaar).	100 kg shoot	100 per pkt	10000.00
Bamboo (culm) (<i>D.Hamiltonnii</i>)		3,000 culm for construction areas.	20 per culm	60,000.00
mushroom	Goling/ Gelephu/ Panbang	100pkt	150/pkt	15000/-
Yula(Microcalamus Andropogonifolius)	Bjoka,	100loads	450/loads	45000/-
Cane shoots	Bjoka	20backloads	300/bk.loads	60000/-
			Total amount	190000/

Table 4: Following prices of finished goods/products emerged from the meeting as below:

Sl. No.	Particulars	Price
1	Bangchung Dagangma	700
2	Bangchung Tangkama	600
3	Bangchung Neykema	350
4	Bangchung Bata barma	250
5	Bangchung Bata	150
6	Baikur(one Pair)big	500
7	Dhungshoop	4000
8	Jaling shoop	3000
9	Tray	1500
10	dhaba	650
11	Zem medium	3000
12	Tangloop	300

9. Procedures to collect the products from the members for sale

- All the members are assigned the products to be weaved (refer to annex.4)

- At the end of the month, products will be collected by the committee members.

10. Marketing Arrangements

The products will be carried to the road point by the members. Products like cane shoots and tsharzaw products in smaller quantities can be sold in *the Pantang market and local festivals*. And when it is collected in larger quantities, it can be sold in *Gelephu/Panbang /KNC* market and to other tourism agents. To confirm the smooth functioning of the business, the dealer from the market will be arranged by the committee members of the group. The chairman and forest officials will monitor and explore the demand from nearby institutes, agents, companies, and other business groups. For further arrangements, support will be sought from the DAMC. In addition, the group will focus more on diversified handicraft development.

11. Benefit to the members

All the members will earn income right after the sale of the NWFPs up to 80% every time and they will deposit 20% in their group account. The 20% of income will be deposited in the bank for 3 years and after that, they will share the annual interest, among themselves. The entire member will have equal rights to collect the resources ensuring future sustainability.

12. By-laws: Group Regulation and Norms:

Members of Tsharzaw group	Village	No. of members households	Geog
	Pongchula	16	Phangkhar
	<u>Responsibilities and rights of members:</u> • Implement activities approved by the government that has been planned in their management plan. • Attain ad-hoc meetings conveyed by committee members. • Protect illegal activities and retain from doing illegal activities in the collection area. • Harvest the bamboo and NWFP only as per the management plan. • Make the products monthly as assigned with particular items.		

Executive committee	
	<u>Roles and responsibilities of committee members:</u>
	• Planning of activities for the NWFP management group. • Active participation in the implementation of group activities. • Ensuring equal share of resources within the group.

Co-ordinator	• Maintain transparency of Products sold by each member and the incentive contributed by each member to the committee members. • Acceptance of new members in the group. • Maintain proper record of group activities, for auditing and monitoring purposes. Name of Co-ordinator: Tshering Yeshi <u>Responsibilities of co-ordinator:</u> • Will explore the marketing of bamboo resources. • Will Plan activities for the user group. • Coordinate activities for the management of NWFP with the group and other stakeholders. • Coordinate and conduct meeting with committee members for the discussion of points with the general user group members. • Submit group monthly activities report to the nearest range office
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Executive committee	
Asst.Co-ordinator	Name of Asst.Co-ordinator: Sangay Penjor <u>Responsibilities:</u> ▪ Obtain a collection permit for the product from the Range office. ▪ Officiate as a chairperson in his absence. ▪ Assist the chairperson in the planning of activities. ▪ Maintain records of any activities conducted in the group. ▪ Maintain a record of tools and implement office correspondence.

Treasurer	Name of Treasurer: Pema Lhadon <u>Responsibilities of Treasurer:</u> • To maintain records of cash transactions. • Collect fines and maintain records of all offences. • Maintain attendance record of meetings and labour works. • Maintain a record of expenditures made for carrying out any activities in the group. • Issue a receipt for the fines collected from the offender.
Compensation to the executive committee for marketing;	The executive members will be refunded for their expenses incurred while going for any official work related to the group.

Meeting	To plan and discuss an important activity for the management of NWFP, meetings will be conducted as follows.		
	Table 5: Meeting schedules:		
	Types of meeting	When	Purpose
	General Meeting	Before going for NWFP harvesting	To know what to do? harvest from where/ and others
	Ad-hoc meeting	Whenever is important	Plan activities as well as prepare activity progress reports.

Membership fee and purpose;	<u>Membership arrangement:</u>		
	- All the households residing in Pongchula chiwog are eligible to join as members of the management group but not before 5 years - Acceptance in the group will be made only by the committee members through the general assembly.		
	<u>Procedure for becoming new members:</u>		
	- Any individual wishing to become a member before the 5-year term is not allowed. - Members should be 18 years old to become a member; - If the re-signee member in future wishes to rejoin the group, he or she should pay Nu of 5000 as a membership fee and get an endorsement from the general meeting of the group.		
	<u>Procedure for leaving the group:</u>		
	- If anyone from the group wishes to discontinue a group member, he/she has the right to leave. However, he/she should leave with the submission of the application to the chairperson and will not allow joining the group in the future. Further, the member leaving the group before completing 5 years will not be allowed benefits from the group fund, and on top of he has to pay Nu5000/-as penalty. - If the children of he/she wish to join as a member in the future, they have to follow the procedure set to become a new member; - If in case the members have a genuine problem and have to leave the group, the executive members will evaluate the		

	problem for confirmation and then they can leave the group with their share.
Formation of NWFP management committee	<u>NWFP Management Committee Formation:</u> NWFP management committee consists of 3 members for <i>Pongchula chewog</i>. The members are elected from the general user group members. <u>Criteria for the selection of committee members:</u> - Any member of the user group has the right to become a committee member. However, they should meet the following requirement: - Should be literate enough to maintain a record (at least in Dzongkha). - Should have leadership qualities to work within the group. - Should have experience in dealing with the outsider

Community development fund and utilization of fund	- The collection of products will be done in a group as per the prescribed collection season; - The 20% fund will be deposited in the group's account and the 80% remaining shall be given to the individuals during the sale of the products. - The group account will also open to deposit the common fund and the account will be operated by the committee members jointly.
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Conflict resolution	- If any disputes arise among the group members the matter will be solved by committee members through meetings; - In case the matter cannot be solved by the committee members the matter will be forwarded to village Tshogpa ; - Still, if it cannot be solved the matter will be further forwarded to the geog administration for final decision.
Penalties	- Absentees while meeting shall be fined Nu: 215/- each time; - For the sick person absentees, the coordinator should monitor him/her.

	• Violation of the NWFP plan shall be fined as per the FNCR and has to pay as per NRPC rate. • The members who couldn't produce the Products as per the task assigned, at the end of the month, will be liable to pay the double price of the products. • If members and others harvest the resources without any valid permit, they will deal as per the Rules and acts of the Forest, Bhutan.
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13. Powers of the Department:

The Department has the right to terminate the NWFP management and marketing plan in case of any violation of the management plan prescription or in the event of inability to carry out the activities as per the plan.

Annex 1 – Membership List

Updated Membership list of Pongchula Tsharzew Tshogpa

Sl.no	Name	CID number	Village	H/H number	sign
1.	Jambay	12006001875	Samdhar	Wa-6-49	
2.	Yesui Dorji	12006001158	Bethangthala	Wa-6-74	
3.	Tendrel Zangmo	12006001174	Pongchula	Wa-6-70	
4.	thechi Dama	12006001051	-u-	Wa-6-68	
5.	pema khaden	11805000715	Gyijong	Wa-6-63	
6.	Dorji choden	12006001187	-u-	Wa-6-63	
7.	Ugyen Ishomo	12006000957	-u-	Wa-6-57	
8.	Sangay Penjor	12006001061	-u-	Wa-6-66	
9.	Yesui Khamo	12006001184	Redijong	Wa-6-76	
10.	Juden	12006001901	-u-	Wa-6-50	
11.	Kinzang	12006001876	-u-	Wa-6-50	
12.	Tshering Yesui	12006001859	Redijong	Wa-6-47	
13.	phuntshe	12006001972	Pongchula	Wa-6-55	
14.	Kinley Wangchuk	12006000996	Gyijong	Wa-6-61	
15.	Tashi choden	12006000175	chabgone	Wa-6-42	
16.	Tandin Wangmo	2006001915	Tradijong	Wa-6-64	

Annex 2 - Result and data analysis of the Yula and Bamboo inventory

Block No.	Village (Blocks)	Subblock No.	Collection Area (Sub-block)		Sample plots		Number of Yula culms to be harvested this year			Number of Yula culms harvested last year			The average number of culms per backload	Range of AHL of Yula backloads (e/i - h/i)
			Name	Area (acres)	Number	Total area (acres)	in plots	per acre (c/b)	in sub-block (d*a)	in plots	per acre (f/b)	in sub-block (g*a)		
				(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
1	Pongchula	1.1	Dhimla	28.8	13	0.08031	154	1917.6	55227.1	5	62.3	1793.1	20	89.7
		1.2	Roksar	5.25	7	0.04324	111	2566.9	13476.2	3	69.4	364.2	20	18.2
		1.3	Charsapong	6.85	8	0.04942	76	1537.8	10534.1	8	161.9	1108.9	20	55.4
		1.4	Redibrang	16.2	9	0.05560	138	2482.1	40210.0	7	125.9	2039.6	20	102.0
		1.5	Manling	2.55	5	0.03089	106	3431.8	8751.0	2	64.8	165.1	20	8.3

1acre 4046.9

40% coordinate subtract steep slopes

Annex 3 - Products assigned to be woven by Members

Sl.no	Name	Products assigned	Numbers per month
1	Jambay	Dhungshoop, Jaling shoop, Tray	One pair
2	Yeshe Dorji	Dhungshoop, Jaling shoop, Tray	One pair
3	Sangay Penjore	Bangchung dhapa	5 numbers
4	Phuntsho	Bangchung Dhapa	5 numbers
5	Kinzang	Zem	1
6	Tshering Yeshe	Zem and Bangchung	1
7	Yuden	Bangchung	5 numbers
8	Yeshe Lhamo	Bangchung	5 numbers
9	Tendrel Zangmo	Bangchung	5 numbers
10	Lachi Dema	Bangchung	5 numbers
11	Jampel Nidup(pema Lhadon)	Bangchung	5 numbers
12	Ugyen Tshomo	Bangchung	5 numbers
13	Kinley wangchuk	Bangchung	5 numbers
14	Tandin Wangmo	Bangchung	5 numbers
15	Tashi Choden	Bangchung	5 numbers
16	Thinley	Tangloop and Ceray	5 numbers

Annex 4 - Yula and Cane resources assessment sheets

Kharila	Sub-Block size(acre)	23.3	Sampling area (acre)	0.2
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The time period for harvesting canes	Sub-block (sub block size = a; sampling area covered = b)			
	Number of canes in sampling area (c)	Number of canes per acre (d) = (c/b)	AHL of cane stems per sub-block (e) = (a*d)	AHL of backloads per sub-block (f) = (e/ average number of canes per backload)
Year 1-2 Present estimated length: >= 15m	15	75	1747.5	200.34
Year 3-4 Present estimated length: 10 - 14.9m	10	50	1165	83.475
Year 5-6 Present estimated length: 6-9.9m	12	60	33.39	16.695
Year 7-8 Present estimated length: 3-5.9m	15	75	100.17	50.085
Year 9-10 Present estimated length: 1-2.9m	9	45	267.12	133.56
Total	29		968.31	484.155

Form I: Yula stock inventory form

Date: 18/11/2022 **Recorder:** PZ **Transect no:** 1
Village: Pongchula **Block:** Pongchula **Sub-block:** Dhimla

Area(acre): 28.8

Coordinates; starting point of sample point line:

Coordinates; End point of sample point line:

Plot No.	Slope (%)	Clump no.	Clump condition*	Total number of culms	Culms to be harvested this year	Culms harvested last year	Estimated number of culms in 1 back-load	Associate species
1	71	4	good	15	7	2	20	
2	82	5	good	17	9	0	20	
3	90	2	poor	7	2	0	20	
4	30	4	good	93	31	1	20	
5	17	3	poor	32	4	0	20	
6	10	2	medium	27	6	1	20	
7	70	0		0	0	0	20	calamus spp.
8	65	3	good	24	15	0	20	calamus spp.
9	62.5	4	good	62	21	0	20	calamus spp.
10	50	2	good	50	20	0	20	
11	77	6	Congested	55	15	0	20	
12	91	2	Congested	45	14	0	20	
13	83	3	Congested	35	10	1	20	calamus spp.

	Total	462	154	5		
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Clump condition*: Good, Medium, Poor, Damaged, or Congested

Clumps to be harvested this year: Number of culms that are 1 or 2 years old (having a second branch at the top) while leaving 2-3 culms in the clump

Form I: Yula stock inventory form

Date: 18/11/2022

Recorder: PZ

Transect no:

Village: Gujong

Block: Pongchula

Sub-block: Roksar

Area(ac): 5.25

Coordinates; starting point of sample point line:

Coordinates; End point of sample point line:

Plot No.	Slope (%)	Clump no.	Clump condition*	Total number of culms	Culms to be harvested this year	Culms harvested last year	Estimated number of culms in 1 back-load
1	88	3	good	30	13	2	20
2	90	5	good	41	15	1	20
3	80	2	congested	44	10	0	20
4	56	5	good	53	21	0	20
5	82	4	good	45	21	0	20
6	91	1	congested	50	5	0	20
7	70	3	good	43	21	0	20

	Total	320	111	3	
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Clump condition*: Good, Medium, Poor, Damaged, or Congested

Clumps to be harvested this year: Number of culms that are 1 or 2 years old (having a second branch at the top) while leaving 2-3 culms in the clump

FormI: Yula stock inventory form

Date: 19/11/2022 **Recorder:** PZ **Transect no:** 1
Village: Redijong **Block:** Pongchula **Sub-block:** Charsapong
area(ac) 6.85
Coordinates; starting point of sample point line:
Coordinates; End point of sample point line:

Plot No.	Slope (%)	Clump no.	Clump condition*	Total number of culms	Culms to be harvested this year	Culms harvested last year	Estimated number of culms in 1 back-load
1	70	3	good	42	20	2	20
2	82	4	good	45	11	1	20
3	71	5	damaged	15	0	2	20
4	80	2	congested	50	5	0	20

5	88	3	good	55	23	1	20
6	80	4	good	41	17	0	20
7	55	2	damaged	13	0	2	20
8	54	3	poor	20	0	0	20
Total				281	76	8	

Clump condition*: Good, Medium, Poor,
Damaged, or Congested

Clumps to be harvested this year: Number of culms that are 1 or 2 years old (having a second branch at the top) while leaving 2-3 culms in the clump

Form I: Yula stock inventory form

Date: 19/11/2022

Recorder: PZ

Transect no:

Village: Redijong
Area(ac) 16.2

Block: Pongchula

Sub-block: Redibrang

Coordinates; starting point of sample point line:

Coordinates; End point of sample point line:

Plot No.	Slope (%)	Clump no.	Clump condition*	Total number of culms	Culms to be harvested this year	Culms harvested last year	Estimated number of culms in 1 back-load

1	89	3	congested	56	8	0	20
2	90	5	good	45	20	2	20
3	75	6	good	51	23	1	20
4	88	3	congested	65	10	1	20
5	91	2	congested	60	15	0	20
6	99	3	good	41	24	1	20
7	50	5	damaged	14	1	0	20
8	77	6	poor	30	7	1	20
9	78	3	congested	50	30	1	20
	Total			412	138	7	

Clump condition*: Good, Medium, Poor, Damaged, or Congested

Clumps to be harvested this year: Number of culms that are 1 or 2 years old (having a second branch at the top) while leaving 2-3 culms in the clump

Form

I: Yula stock inventory form

Date: 19/11/2022

Recorder: PZ

Transect no: 1

Village: Redijong
Area 2.55

Block: Pongchula

Sub-block: Manling

Coordinates; starting point of sample point line:

Coordinates; End point of sample point line:

Plot No.	Slope (%)	Clump no.	Clump condition*	Total number of culms	Culms to be harvested this year	Culms harvested last year	Estimated number of culms in 1 back-load
1	91	2	poor	21	6	0	20
2	89	4	congested	77	34	1	20
3	99	5	congested	59	30	0	20
4	88	6	poor	18	7	0	20
5	90	5	congested	58	29	1	20
	Total			233	106	2	

Clump condition*: Good, Medium, Poor, Damaged, or Congested

Clumps to be harvested this year: Number of culms that are 1 or 2 years old (having a second branch at the top) while leaving 2-3 culms in the clump